# Monotonicity: Formula Sheet

## Definitions

Increasing at $a$: $f(a-h) < f(a) < f(a+h)$

Decreasing at $a$: $f(a-h) > f(a) > f(a+h)$

Increasing on $I$: $x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$

Non-decreasing on $I$:

$$x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$$

Decreasing on $I$: $x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$

Non-increasing on $I$:

$$x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$$

## Derivative tests

Increasing: $f'(x) \geq 0$, not $\equiv 0$ on any interval

Decreasing: $f'(x) \leq 0$, not $\equiv 0$ on any interval

Composite: $(f \circ g)' = f'(g(x))\, g'(x)$

Critical point: $f'(c) = 0$ or $f'(c)$ does not exist

Stationary point: $f'(c) = 0$

## Always increasing / decreasing

Cubic monotonic: $b^2 - 3ac \leq 0$

$ax^2 + bx + c \geq 0\ \forall x$: $a > 0,\ D \leq 0$

$ax - \sin x$ increasing: $a \geq 1$; decreasing: $a \leq -1$

Separated parameter: $p \geq \max g$ (increasing), $p \leq \min g$ (decreasing)

## Values, comparison, inequalities

$\left(\frac{\ln x}{x}\right)' = \frac{1-\ln x}{x^2}$; $e \leq a < b \Rightarrow a^b > b^a$

$e^\pi > \pi^e$; $3^4 > 4^3$

$x > \sin x$ and $\tan x > x$ for $0 < x < \frac{\pi}{2}$

$\frac{x}{1+x} < \ln(1+x) < x$ for $x > 0$

$e^x > 1 + x$ for $x \neq 0$

$f \uparrow$: $f(A) > f(B) \iff A > B$; $f \downarrow$: $\iff A < B$

## Rolle's theorem

Conditions: continuous $[a, b]$, differentiable $(a, b)$, $f(a) = f(b)$

Result: $f'(c) = 0$, $a < c < b$

Roots: $n$ of $f \Rightarrow \geq n - 1$ of $f'$

Roots: $m$ of $f' \Rightarrow \leq m + 1$ of $f$

$f'(c) + kf(c) = 0$: use $e^{kx} f(x)$

$cf'(c) = f(c)$: use $\frac{f(x)}{x}$

$f'(c) = k\phi'(c)$: use $f - k\phi$

## LMVT

$f'(c) = \frac{f(b)-f(a)}{b-a}$, $a < c < b$

$f(a+h) = f(a) + hf'(a+\theta h)$, $0 < \theta < 1$

Proof function: $g = f - \frac{f(b)-f(a)}{b-a}x$

Quadratic: $c = \frac{a+b}{2}$

Bounds: $m(b-a) \leq f(b) - f(a) \leq M(b-a)$

## Asymptotes

Vertical:

$$\lim_{x \to a^\pm} f = \pm\infty$$

Horizontal: $\lim_{x \to \pm\infty} f = a$

Inclined: $m = \lim \frac{f}{x}$, $b = \lim(f - mx)$

Rational, $n < m$: $y = 0$; $n = m$: $y = \frac{a_0}{b_0}$

Rational, $n = m + 1$: quotient $ax + b$